

JAY ZHAO

zhaoj06@bu.edu | (518) 812-5682 | www.linkedin.com/in/jzhaonys | Boston, MA

EDUCATION

Boston University, College of Engineering
Bachelor of Science, Mechanical Engineering

Boston, MA
Expected May 2028

- GPA: 3.3/4.0
- Relevant Coursework: Hands-on Engineering, Introduction to Engineering Design, CAD and Machine Components, Engineering Mechanics, Programming for Engineers (MATLAB, Python)
- Activities & Societies: Terrier Motorsports (Formula SAE), Sigma Chi Fraternity, Music Business Club, EK100 Freshmen Student Advisor

SKILLS

Software: SolidWorks, OnShape, Autodesk Fusion, MATLAB, Python, Cura, Microsoft Office, Google Suite

Technical: CAD modeling, Soldering, 3D-printing, Wiring/Circuit Assembly, Laser Cutting, Manual Mill/Lathe

Interpersonal: Positive Influence/Leadership in Team-Based Environments, Attention to Detail, Flexibility

EXPERIENCE

Engineering Intern

Jun 2026 - Aug 2026

Becton Dickinson (BD)

Glens Falls, NY

- Designed 7 custom fixtures, tools, and machine guarding solutions to support catheter manufacturing operations, utilizing SolidWorks and 3D printing to prototype and validate designs
- Drafted and presented an acceptance criteria-change project to division leadership, identifying an opportunity with an estimated \$50K in savings and \$15K in annual recurring savings
- Revamped an existing spray rack fixture by redesigning gear train system and creating a new overspray shield, reducing component repair costs by 90% and contributing to \$135K in projected annual repair labor savings
- Supported 3 Kaizen events by performing data collection and analysis on product output and cycle times

Wait Staff

May 2025 - May 2026

Thai Sushi Aroy Dee

Glens Falls, NY

- Processed 75+ customer orders per night using a digital POS system, executed opening and closing procedures, including setup, sanitation, and equipment security, to maintain efficient service and safe operations

PROJECTS

Automatic Color Sorter

Feb 2026 - Apr 2026

- Collaborated with 3 peers to design and develop an Arduino-based color sorting system capable of identifying and separating colored candies at an accuracy rate >90%, and under a budget of \$200
- Modeled a modular housing system in SolidWorks, manufacturing parts via 3D-printing and laser cutting MDF, assembled and wired electronic systems
- Developed and iteratively refined test code to calibrate color sensor and motor, improving actuation accuracy while preventing candies from escaping system, resulting in a color identification rate of 100% an accuracy rate of 92% and a total cost of \$56

DC Motor Car, Terrier Motorsports Onboarding Project

Feb 2026 - Apr 2026

- Tasked to design, develop and manufacture a 9V DC motor powered car within a 120mm × 120mm chassis constraint, powered by a DC motor and capable of traveling across a table under 10 seconds as part of onboarding process
- Designed and CAD-modeled a full vehicle assembly in Autodesk Fusion, including chassis, motor mount, aerodynamic cover, wheels, and battery mount, and machined parts out of HDPE and acrylic stock

Room Temperature Monitor Box

Feb 2025 - Apr 2025

- Created an Arduino-based ambient temperature monitor using a TMP36 sensor and LCD to display real-time data, with a PKM17 buzzer for alerts above 90°F
- Modeled a custom 9V battery holder and enclosure in Onshape and conducted CAD-based assembly of components to validate fit, tolerances, and system integration before 3D-printing parts
- Programmed an algorithm to acquire and process TMP36 sensor data, testing and refining the code in order to improve measurement accuracy and overall system performance